

There are many problems within fluid mechanics, computational chemistry, and material science that are intractable to solve using classical computers. For instance, simulating fluid flow at high fidelity requires extensive computational resources that are either unavailable or would require years of computation time. On the other hand, some of these problems may map well to quantum computers, enabling us to tackle a new range of problems. For example, quantum computing can transform materials science problems by using quantum mechanics to directly simulate atomic and molecular interactions, and the Quantum Lattice-Boltzman Method can model fluid behavior via particle movement and collision steps on quantum circuits.

However, there are still many challenges to making quantum computing practical, including overcoming noisy devices, high error correction overhead, and interconnection bottlenecks. Quantum algorithms must be designed to overcome these challenges.

Exam Tasks

For this qualifying exam, you will be investigating quantum simulation algorithms and techniques for analyzing fluid flow. You are required to complete two main tasks:

1. Conduct a comprehensive literature survey of simulation methods for analyzing fluid flow in computational fluid dynamics. This survey should include techniques and algorithms using classical computing systems as well as those relying on quantum computing paradigms.
2. Propose a new (or improved upon) quantum algorithm for simulating fluid flow.

Literature Survey Questions

Your survey should address (but not be limited to) the following questions:

- What are the state-of-the-art classical computing techniques used for simulating fluid flows and what are their inherent limitations and challenges? This investigation must include an extensive literature review and should consider relevant work over the past 15 years.
- What are the inherent properties of quantum computing that provide clear advantages over classical computing techniques? What are the physical challenges that limit quantum computing from becoming practical today?

- What properties of fluid flow make it particularly suitable for quantum simulation? What significance may it have for fluid dynamics research? What are the state-of-the-art proposed quantum computing techniques to simulate fluid flow?
- How do the current physical limitations in quantum computing impact the ability to accurately and practically simulate fluid flow using a quantum paradigm? How may a hybrid classical-quantum approach be used to address some of these limitations?

Research Project Details

You are expected to design and analyze a new (or improved upon) quantum computational fluid dynamics algorithm. While full implementation of this algorithm may not be possible by exam time, an analysis of the properties and complexity of the algorithm will be expected.

Your project must include the following components:

- Provide a detailed description of a specific quantum computational fluid dynamics algorithm. What are the main challenges in implementing this algorithm on a quantum computer?
- What strategies have been used to overcome these challenges for a realizable quantum implementation? Discuss how machine learning (or other complementary computing technique) may be used to help with these challenges.
- Propose a new algorithmic strategy that addresses at least one of the challenges. Justify your strategy. Consider how new advances in quantum technology may enable your strategy to become more practical or advantageous.
- Provide a prototype of your algorithm. How would you go about fully designing and testing your algorithm? Which existing quantum software tools and platforms would you use and why?

Deliverables:

1. A conference-quality research paper. The paper should be approximately 8-12 pages in length and include an introduction and motivation, background and related work, algorithm design, detailed strategy for full implementation and testing of the algorithm, discussion and future directions.
2. A 30-minute oral presentation of your algorithm to the qualifying exam committee followed by questions and answers.